## 1 Query-conditioned candidate pool



**Query**

*"Locate the handle of the mug."*

$\phi(x)$ = { "the handle of the mug", "the mug" }

SAM 3 is prompted with the query's phrases

**SAM 3 · point grid**
query-free, 32×32 points

**SAM 3 · text**
top-3 instances per phrase

$\mathcal{V} = \mathcal{G} \cup \mathcal{T}$ one pool per query




$\mathcal{G}$ grid masks

$\mathcal{T}$ SAM 3 text instances

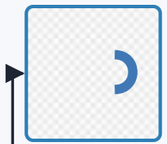
**Containment tree**

$$\text{parent}(v) = \underset{u\,:\,|m_u \cap m_v| \geq 0.8\,|m_v|}{\arg\min} |m_u|$$

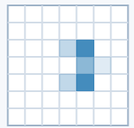
every candidate links to its smallest container;
the relation of two candidates biases attention (panel 3)



## 2 One token per candidate



$m_v$



$c_v$

coverage of each image-token cell

**appearance**
$p_v = \frac{\sum_j c_{v,j}\, e_j}{\sum_j c_{v,j}}$

**geometry**
$g_v \in \mathbb{R}^9$
box, area, depth

**provenance**
$q_v \in \mathbb{R}^5$
text?, rank, score

$$\hat{x}_v = \text{RMSNorm}\big(\text{MLP}([p_v;\, g_v;\, q_v])\big)$$

**Where the token sits and what it sees**

$\text{pos}(\hat{x}_v) = (t_{\text{end}},\, \bar{r}_v,\, \bar{c}_v)$
placed at the mask centroid (mRoPE)

$\mathcal{R}_v = \text{dilate}\big(\text{supp}\, c_v\big)$ image region

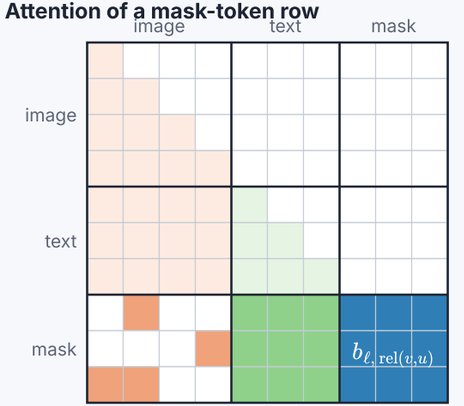
*one forward pass for all candidates*

## 3 Selection inside the LLM




**Qwen3-VL-4B + LoRA (r = 16)**
text and image tokens are unchanged: they never attend to mask tokens

**Attention of a mask-token row**




query text: always visible

image: only its region
$\mathcal{R}_v = \text{dilate}\big(\text{supp}\, c_v\big)$

other candidates: bias by tree relation

blocked

$z_v$

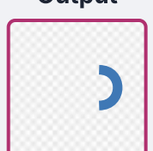
**Selection head**

$$s_v = \lambda\, \cos\big(\psi_q(h_{\text{SEL}}),\, \psi_k(z_v)\big) + \beta$$

$v^\star = \arg\max_{v \in \mathcal{V}}\, s_v$

$\max_v\, \sigma(s_v) < \tau \;\Rightarrow\;$ no target

**Output**



$m_{v^\star}$